

UNIT: 10 Multimedia

Contains:

- ❑ Definition,
- ❑ Characteristics,
- ❑ Elements,
- ❑ Applications, Benefits,
- ❑ Tools,

Introduction to Multimedia

- Multimedia is an interactive media and provides multiple ways to represent information to the user in a powerful manner.
- It provides an interaction between users and digital information.
- It is a medium of communication.
- Some of the sectors where multimedia is used extensively are education, training, reference material, business presentations, advertising and documentaries.

What is Multimedia?

- Multimedia is a representation of information in an attractive and interactive manner with the use of a combination of text, audio, video, graphics and animation.
- In other words we can say that Multimedia is a computerized method of presenting information combining textual data, audio, visuals (video), graphics and animations.
- For examples: E-Mail, Yahoo Messenger, Video Conferencing, and Multimedia Message Service (MMS).
- Multimedia as name suggests is the combination of Multi and Media that is many types of media (hardware/software) used for communication of information.

What is Multimedia? Cont..

- Multimedia is a form of communication that combines different content forms such as text, audio, images, animations, or video into a single presentation, in contrast to traditional mass media, such as printed material or audio recordings.
- **Multimedia** is a broad term for combining multiple media formats.
- Whenever text, audio, still images, animation, video and interactivity are combined together, the result is **multimedia**.
- Slides, for **example**, are **multimedia** as they combine text and images, and sometimes video and other types.

Multimedia System

- Multimedia systems are those computer platforms and software tools that support the interactive uses of text, graphics, animation, audio or motion pictures.
- It is the computer able to handle multiple media, store, digitize, compress, retrieve and decompress the information.
- It includes hardware like CD-ROM, Sound Card such as Sound Blaster or Master Blaster, Microphone, Head Phones, Digital Camera, Speakers, Home Theaters and Multimedia Projectors.

Multimedia System Cont..

- It is not a single technology, but rather a collection of technologies for manipulating
 - Text(characters, digits, symbols)
 - Graphics(images and animation)
 - Audio
 - Video
- It is a medium that includes the four factors:
 - Digitization
 - Networking
 - Interactivity
 - Inclusion of human interfaces

Multimedia Technology

- It is the special computerized technique which helps user by providing methods to combine text, images, and sound or motion pictures.
- It facilitates the user to create, store, edit, delete and copy data.
- Standard file formats are used to create or store multimedia information
- Graphics are stored in .bmp, .jpg, .gif, .tif, .pif, .png formats
- Audio is stored in .wav, .mp3, .wma formats
- Videos and motions pictures are stored in .mpg, .avi, .3gp formats

Software Used In Multimedia

- Multimedia is widely used in web, offices, educational presentations, game or business presentation, entertainment, trainings. Hence, programmers or software developers use a variety of software as per the requirement.
- Java (Programming Language)
- Real Audio Or Shock wave (Utilities)
- Active –X of Open Dock
- Windows Media Player
- Flash, Photoshop, Pagemaker, Dreamweaver, 3D-Max, Maya (Graphics, Designing, Editing)

Elements of Multimedia

Five Elements

- **Text,**
- **image,**
- **audio,**
- **video, and**
- **animation**

Components of Multimedia

- **Text-** All multimedia productions contain some amount of text. The text can have various types of fonts and sizes to suit the professional presentation of the multimedia software.
- **Graphics-** Graphics make the multimedia application attractive
- There are two types of Graphics:
 - **Bitmap images-** Bitmap images are real images that can be captured from devices such as digital cameras or scanners. Generally bitmap images are not editable. Bitmap images require a large amount of memory.
 - **Vector Graphics-** Vector graphics are drawn on the computer and only require a small amount of memory. These graphics are editable.
- **Audio-** A multimedia application may require the use of speech, music and sound effects.

Components of Multimedia Cont..

- **Video-** The term video refers to the moving picture, accompanied by sound such as a picture in television. Video element of multimedia application gives a lot of information in small duration of time.
- **Animation-** Animation is a process of making a static image look like it is moving. An animation is just a continuous series of still images that are displayed in a sequence.
- The animation can be used effectively for attracting attention. Animation also makes a presentation light and attractive.

Characteristics of Multimedia Systems

Four characteristics of multimedia systems

- Multimedia systems must be computer controlled.
- They are integrated.
- The information they handle must be represented digitally.
- The interface to the final user may permit interactivity.

Multimedia Application

- The capability of multimedia technology to handle different type of media makes them suitable for wide range of applications and users.
- It has gained world wide acceptance in almost a business, presentation or simulation education or training entertainment, video, graphics, or animations. Some of the areas where multimedia could have direct impact are
 - Video Games
 - Multimedia Presentation
 - Special Effects in Film
 - Public Accessing/Touch Screen Monitors
 - Animated Advertisement
 - Foreign Language Learning
 - Virtual Reality
 - Edutainment
 - Software Training
 - Internet And Interactive Web Pages
 - Office Work
 - Multimedia Server and databases

Applications of Multimedia Cont..

- **Multimedia in Marketing and Advertising-**
- **Multimedia in Entertainment-**
- **Multimedia in Education-**
- **Multimedia in Bank-**
- **Multimedia in Hospital-**
- **Communication Technology and Multimedia Services-** The advancement of high computing abilities, communication ways and relevant standards has started the beginning of an era where you will be provided with multimedia facilities at home.
- These services may include:
 - Basic Television Services
 - Interactive entertainment
 - Digital Audio
 - Video on demand
 - Home shopping
 - Financial Transactions
 - Interactive multiplayer or single player games
 - Digital multimedia libraries
 - E-Newspapers, e-magazines

Benefits of Multimedia

- Easy to understand and use
- Integrated and interactive
- Conducive to co-operative work environment
- Flexible
- Supportive of large audience

Multimedia Tools

- Photo Editing : Photoshop, Splashup
- Web Design : Dreamweaver, KompoZer
- Video : Final Cut, Adobe Premiere
- Audio : ProTools, Adobe Audition
- Slideshows : Soundslides, PhotoPeach
- Interactive media : Flash, Effect Generator
- Word processing : Microsoft Office, Google Docs

Advantages of Multimedia

- It makes teaching learning easier in the classroom
- It makes sharing of views, ideas and thoughts among various people around the world easy.
- It can store the data and information for long time.
- It is very cheap to get the knowledge about the related subject matter in a short time through multimedia.
- It is very easy to use, handle, carry, copy and store data.
- It allows adding audio, video, text and graphics to make the subject matter interactive and attractive.
- It has a wide use in interactive web pages, video conferencing, distance education and seminars.



Thank you!